



Power Electronics

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Gate Driver & Snubber circuit design



Key points of this topic

- (1) Why we need snubber in practical application?
- (2) Diode snubber;
- (3) Overvoltage snubber for MOSFET and IGBT;
- (4) Over-current protect of power devices.
- (5) Gate driver design for MOSFET and IGBT.
- (6) Impact of driving resistance to the transition curves of voltage and current;

Ref. Chapter 18: Protection of Devices and Circuits
Chapter 17: Gate Drive Circuits



1. Why we need snubber?



(1) di/dt or dv/dt limits

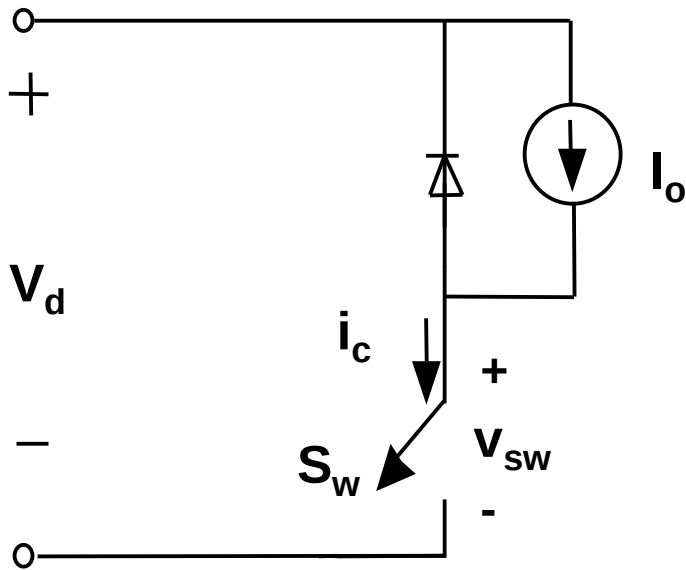
Also, some devices have di/dt or dv/dt limits (esp. thyristor type devices). Higher dv/dt may cause latchup for IGBT and MOSFET

(2) EMI

High di/dt , dv/dt lead to EMI, affects control circuits.
($dv/di \rightarrow$ capacitive coupling, $di/dt \rightarrow$ inductive coupling)



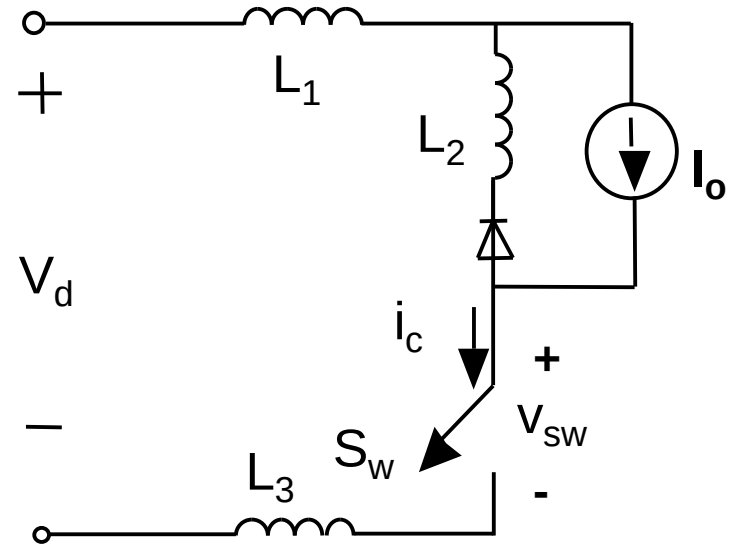
(3) Overshoots



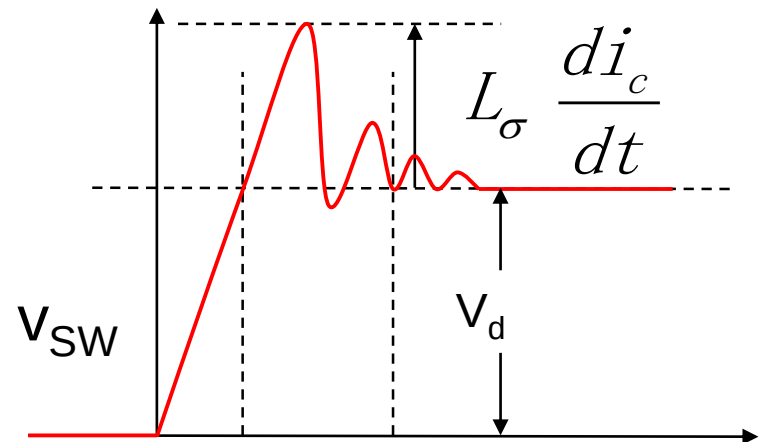
(a) Ideal circuit

L_1, L_2, L_3 are
stray inductor

$$L_\sigma = L_1 + L_2 + L_3$$



(b) Real circuit



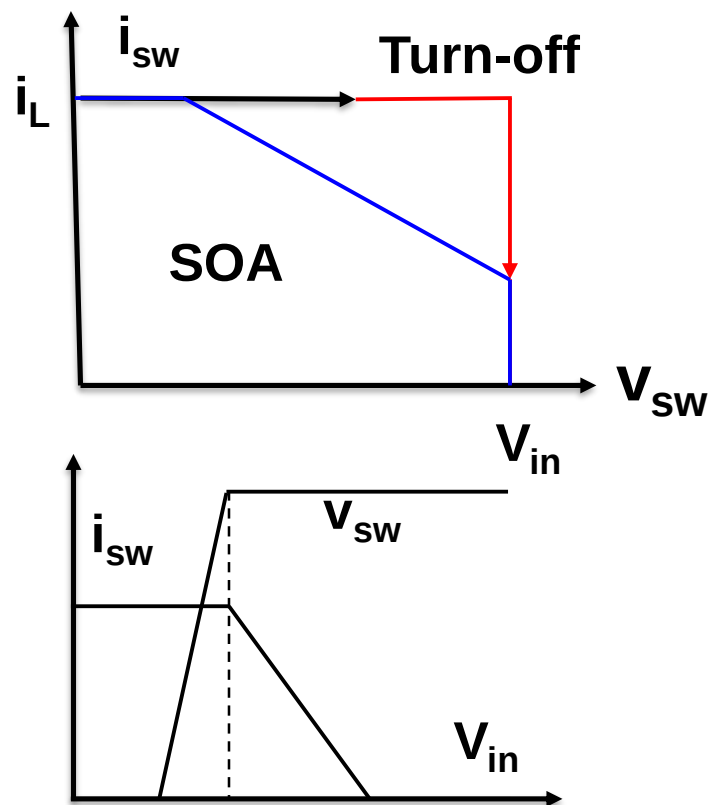


(4) Safe Operating Area (SOA)

PE devices have limitations on simultaneous **voltage** & **current** applied instantaneously (IGBT, MOSFET, etc.)

SOA of MOSFET & IGBT

Device can operate at I_L
and can operate at V_{in} ,
but not at same time





2. Snubber circuit

(1) Snubber's Function is to protect semiconductor devices by

- (a) Limiting device **over voltages** during turn-off transients.
- (b) Limiting the rate-of-rise of voltages (dv/dt) across the semiconductor device at device turn-off .
- (c) Limiting device di/dt through the semiconductor device during turn-on transients.
- (d) Shaping the switching trajectory of the device as it turns on/off and lows the device **power dissipation**

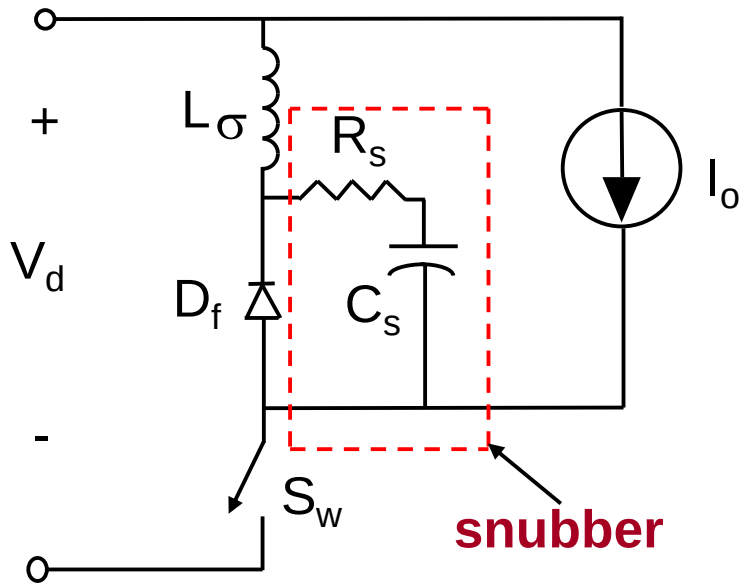
(2) Basic snubber circuit

- (a) R-C snubber, (b) R-C-D snubber, (c) L snubber

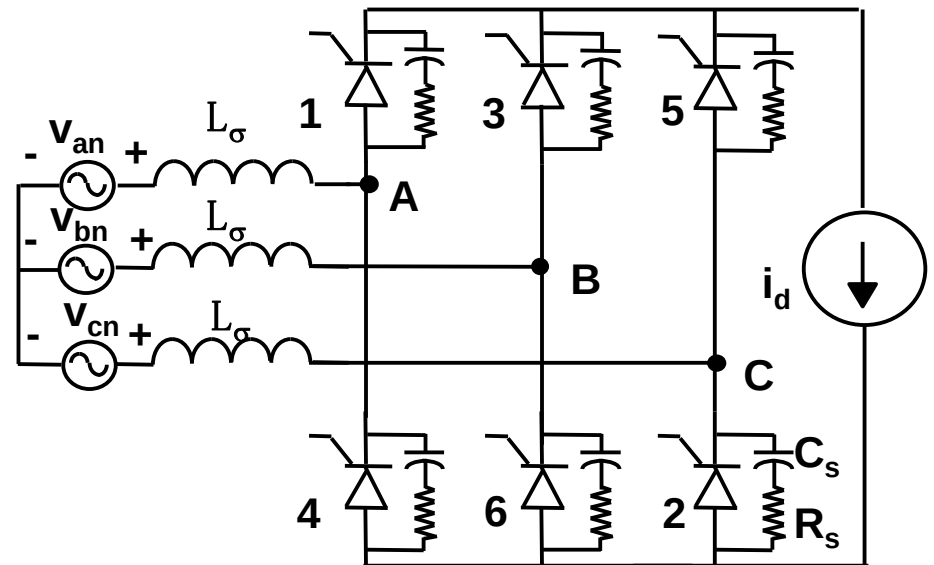


(a) R-C snubber (Unpolarized)

Generally, it is used to protect **Diodes** and **Thyristors** by limiting the peak transient voltage across the devices and dv/dt during reverse recovery time.



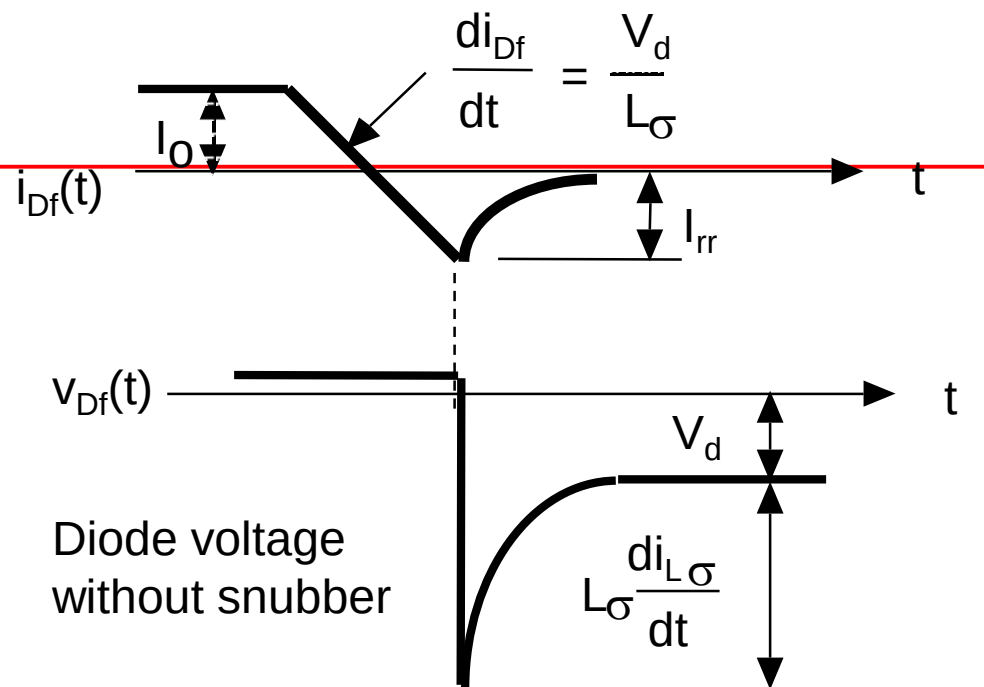
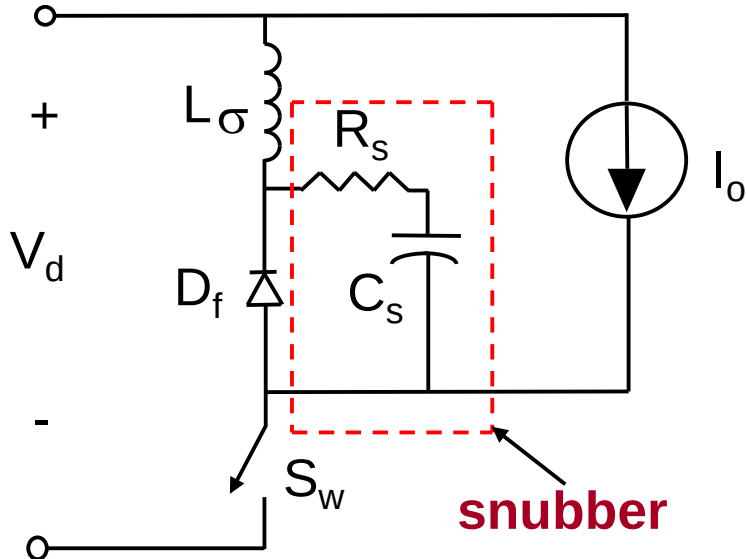
R-C snubber for diode



R-C snubber for thyristor



Introducing a R_s - C_s snubber circuit



$$C_{base} = L_\sigma \left[\frac{I_{rr}}{V_d} \right]^2$$

$$R_{base} = \frac{V_d}{I_{rr}}$$

Notes for R_s - C_s parameters in R-C snubber design:

(1) Generally, both R_s and C_s of diode snubber can be chosen as base value of them respectively.

(2) However, snubber is not necessary in many application due to negligible energy in L_σ because of super layout and excellent characteristics of the employed diodes

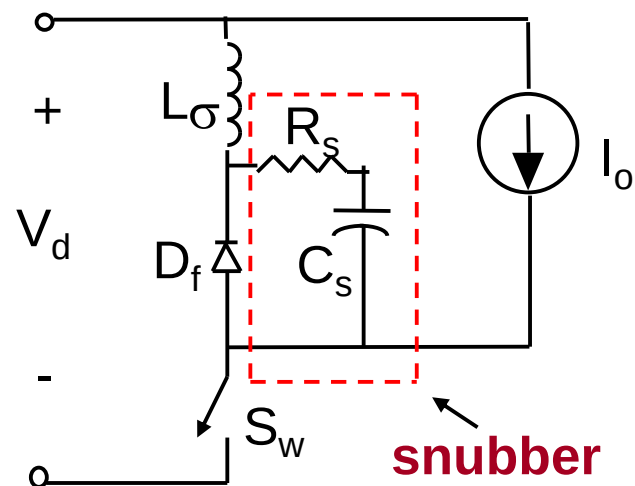


overvoltage protection R_s - C_s snubber circuit

The roles of elements

C_s 的作用是限制 D_f 上的 dv/dt 和关断过电压；

R_s 的作用是限制 C_s 的放电电流。

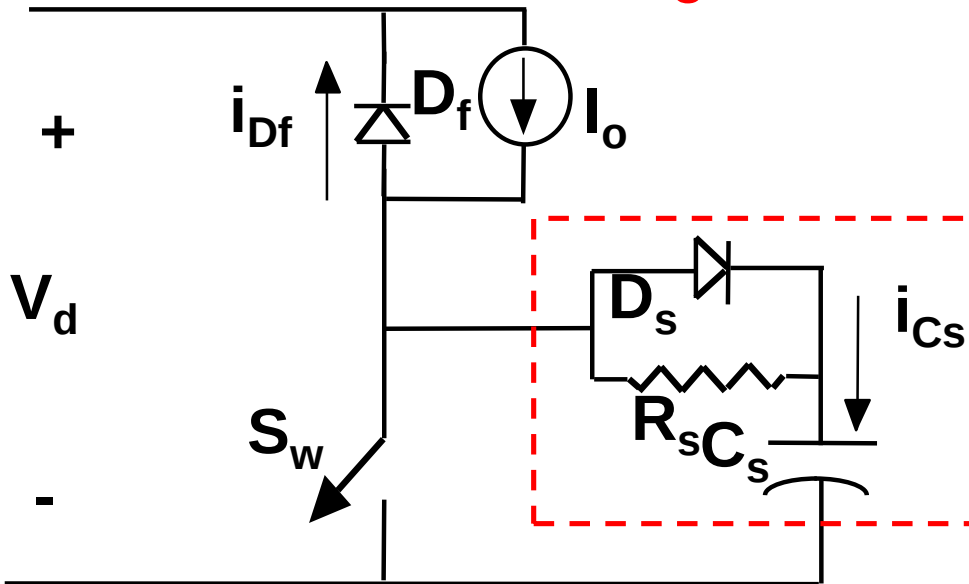




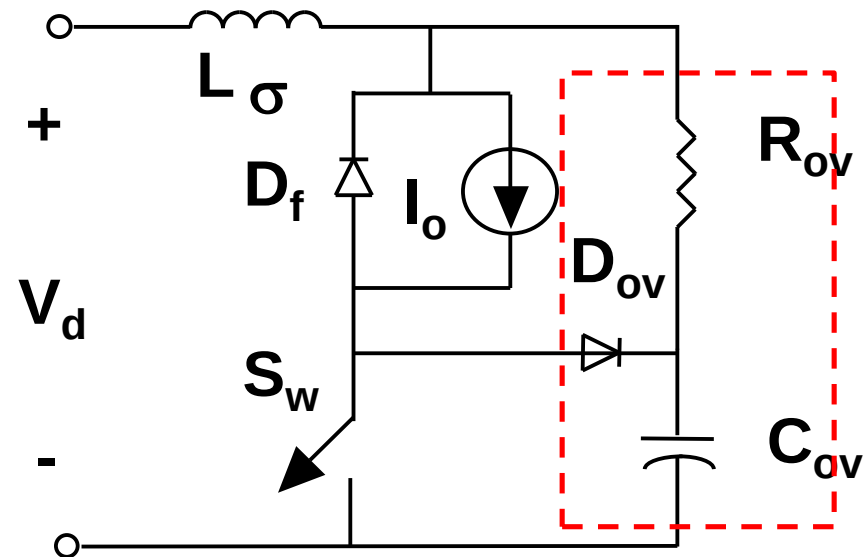
(b) RCD snubbers

- ◆ Used as turn-off snubbers to shape the turn-off switching trajectory of controlled switches.
- ◆ Used as overvoltage snubbers to clamp voltages applied to controlled switches to safe values
- ◆ Limit dv/dt during device turn-off

Used for switching device MOSFET, IGBT, BJT, IGCT etc.



Turn-off snubbers to limit dv/dt



Overvoltage snubbers

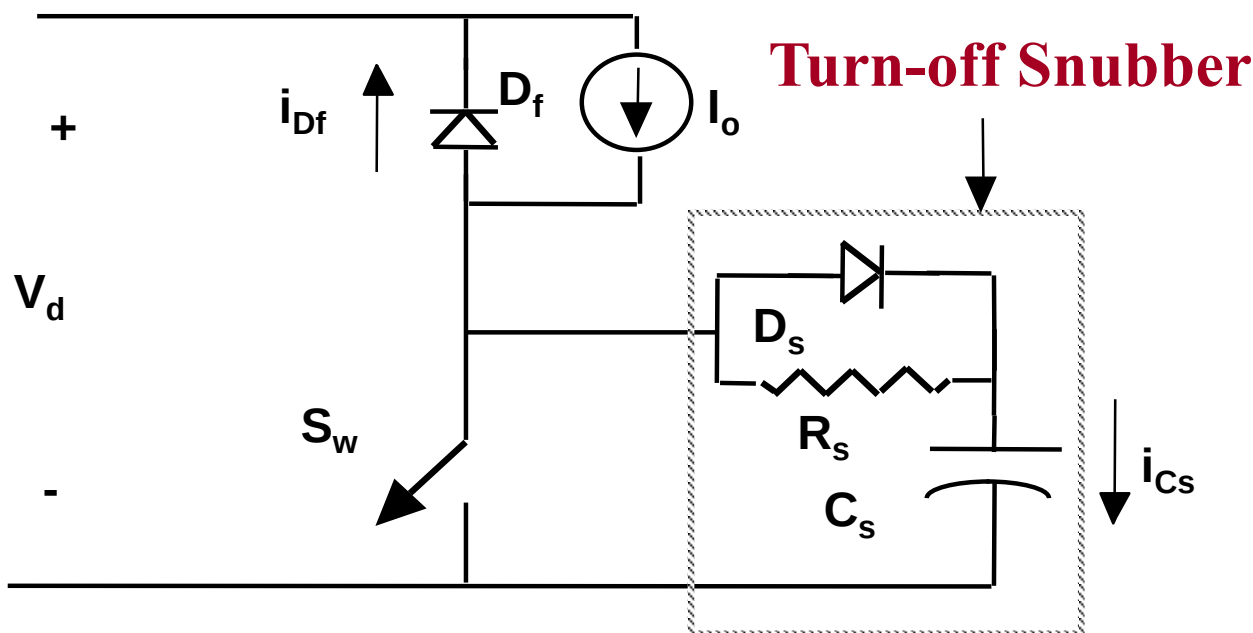


RCD snubber 各元件的作用

C_s 的作用是限制上 S_w 的 dv/dt 和限制关断过电压；

R_s 的作用是当 S_w 开通时泄放 C_s 上的能量，并限制放电电流；

D_s 的作用是当 S_w 关断时使 R_s 短路，更好发挥限制上 S_w 的 dv/dt 和限制关断过电压的作用。





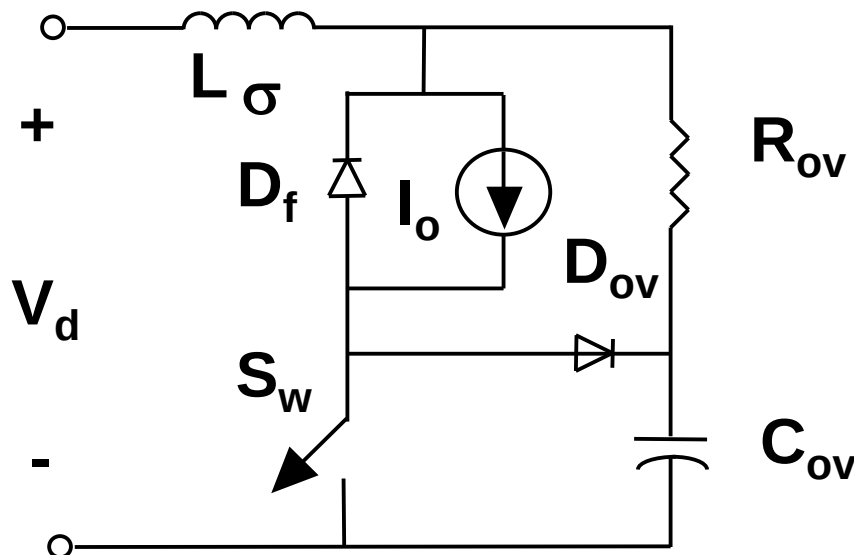
➤ The roles of elements

C_{ov} 的作用是限制 S_w 上的 dv/dt 和限制关断过电压;

R_{ov} 的作用泄放 C_{ov} 上的能量, 并限制放电电流;

D_{ov} 的作用是提供 C_{ov} 充电回路, 更好发挥限制 S_w 的 dv/dt 和限制关断过电压的作用。

Overvoltage Snubber



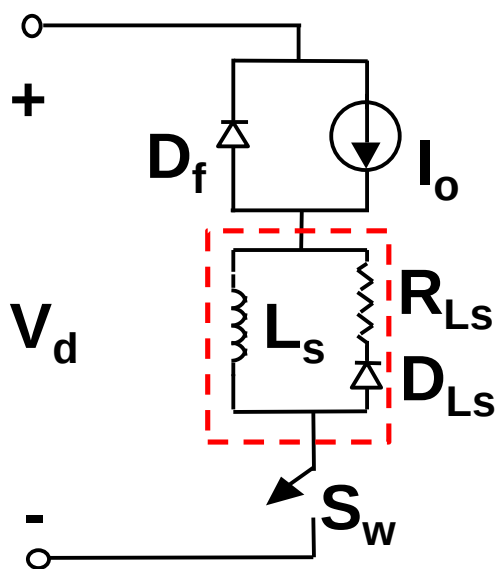


(c) Polarized L-R snubbers

- ◆ Used as turn-on snubbers to shape the turn-on switching trajectory of controlled switches.

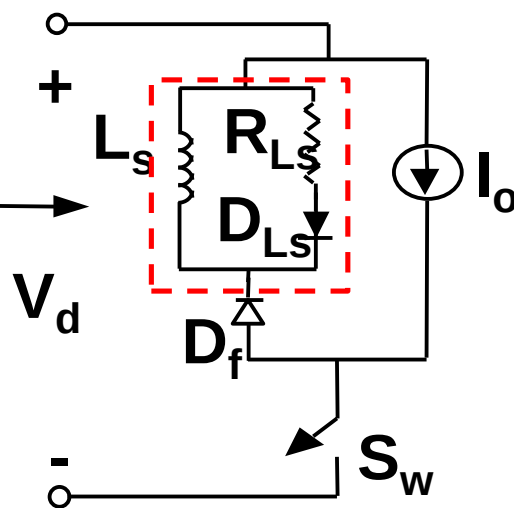
- ◆ Limit di/dt during device turn-on

Especially for slow switching devices, Such as Thyristor, GTO, IGCT



Limit di/dt

Snubber
circuit



Turn-on snubber

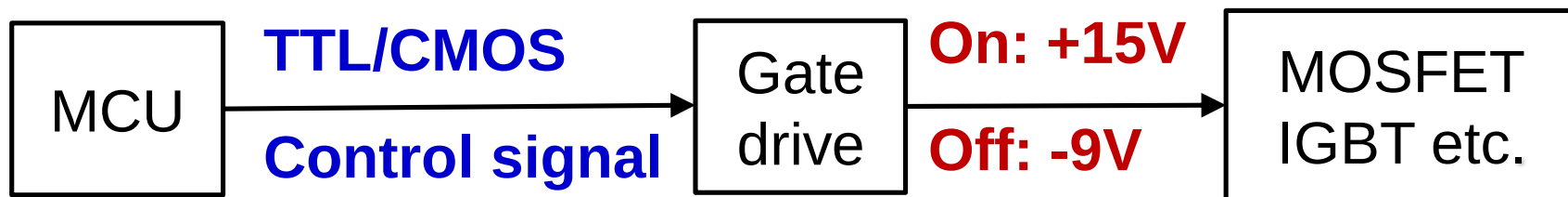


3. Gate Drive Circuits

(1) What function does gate drive circuit have?

a) Provide interface between the control circuit and power switch

- ◆ It **amplifies control signals** to levels required to drive power switch
- ◆ **Provides electrical isolation** between power switch and logic level signal control circuits when it is needed



When electrical isolation is necessary?



b) Waveshaping output of the drive circuit to improve the power switch turn-on or turn-off performance.

- ◆ **Minimize turn-on and turn-off time to reduce switching loss.**
- ◆ **Provide adequate drive power to keep power switch in on-state**
- ◆ **Provide reverse bias to ensure that power switch remains off and is not prone to be triggered by transient noise.**

c) Provides overcurrent or overvoltage protections, blanking time for complementary (互补) devices.



(2) How to design gate drive circuit

Key features for gate drive circuit design consideration:

(1) Drive circuit topologies

Output signal polarity.

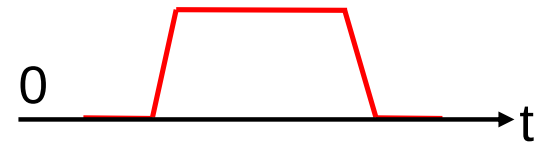
(2) Output peak current capability

Larger I_g speed-up turn-on and turn-off process.

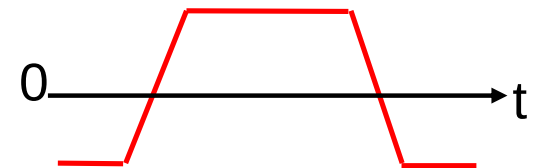
(3) Protection for power switch

- ◆ Overcurrents or short circuit protection
- ◆ Blanking time for complementary devices

(4) Have minimize stray inductance across gate drive wire and be immune to switching noise.



Unipolar



Bipolar



Example1. Bipolar Gate drive circuit for IGBT and MOSFET

Operating principle of the gate drive circuit shown in Fig.28-5(a).

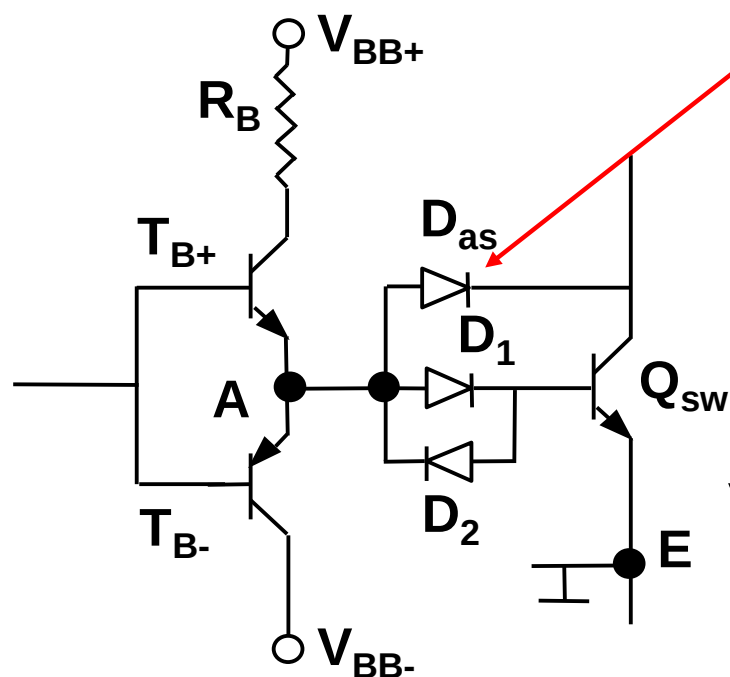


Fig. 28-5 (a)

Anti-saturation diode D_{as} keeps Q_{sw} slightly out of saturation.

$$V_{AE} = V_{BE(on)} + V_{D1} = V_{CE(on)} + V_{Das} \quad (28-6)$$

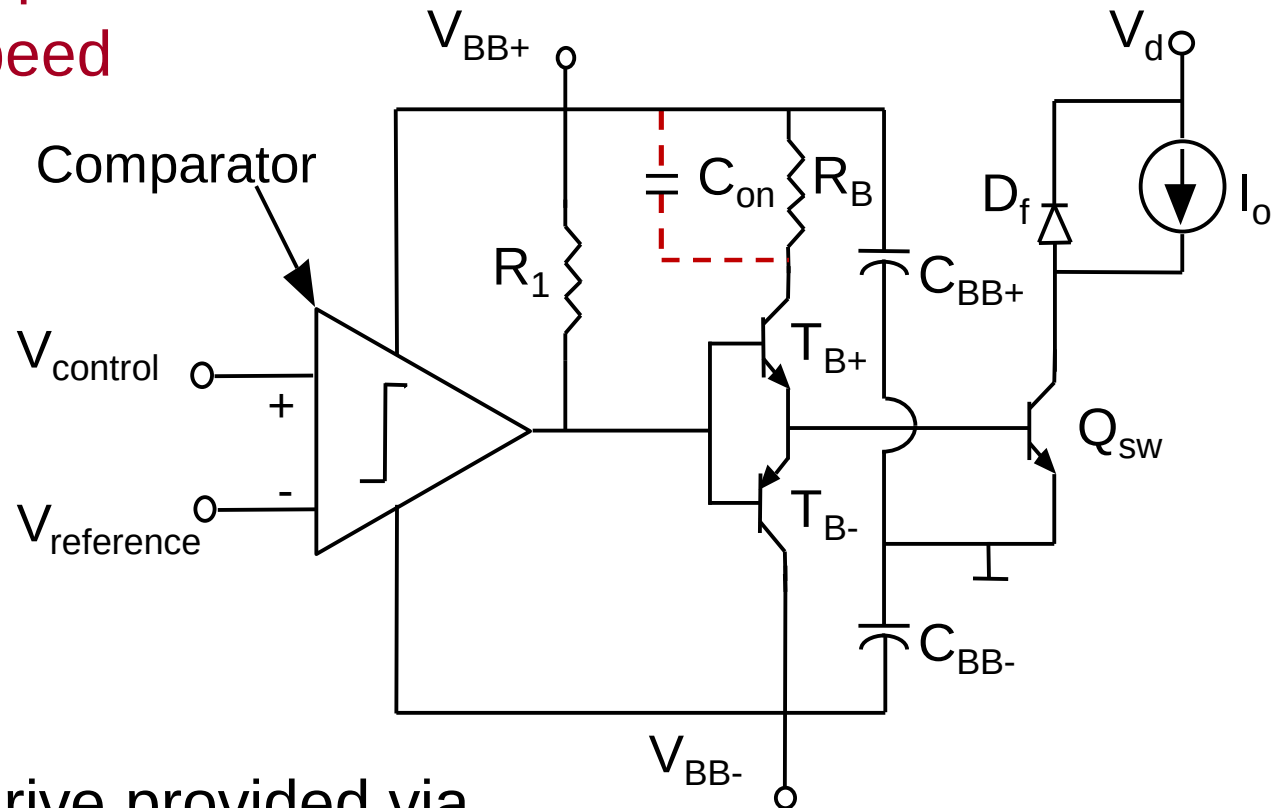
$$V_{CE(on)} = V_{BE(on)} > V_{CE(sat)} \text{ because } V_{D1} = V_{Das} \quad (28-7)$$

D_2 provides path for negative base current at Q_{sw} turn-off.



Example2. Bipolar Gate drive circuit for IGBT and MOSFET

C_{on} ---- speed-up
capacitor to speed
up the turn-on
sequence.

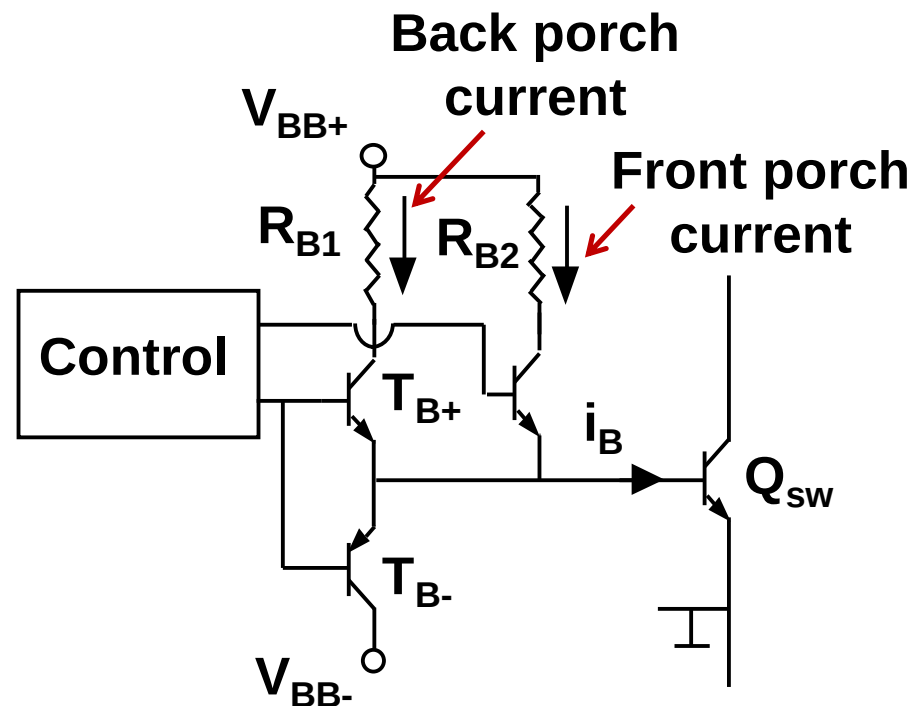
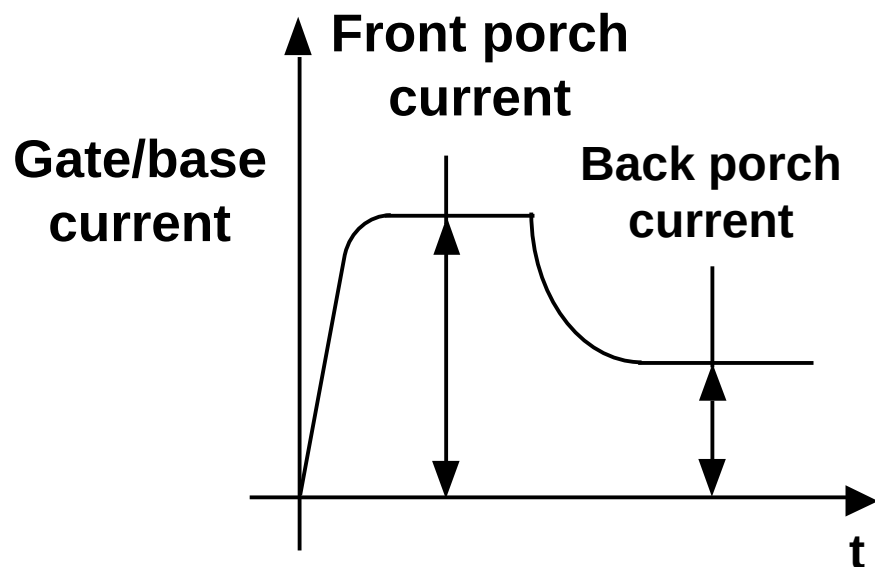


Transient overdrive provided via C_{on} for faster turn-on of switch

Fig.28-4(a)



Example3. Bipolar Gate drive circuit for IGBT and MOSFET



Front porch, back porch gate/base currents at turn-on

Faster turn-on without putting device deeply into on-state where turn-off delay time will be substantially increased.

Applicable to BJTs, MOSFETs, IGBTs, and GTOs.



Example 4. Unipolar Gate Drive Circuit

Unipolar drive circuits are suitable for converters with a single-switch

(a) Simple gate drive circuit

Circuit operation

- $V_{\text{control}} > V_{\text{reference}}$: comparator output high and Q_{sw} on.
- $V_{\text{control}} < V_{\text{reference}}$: comparator output low and Q_{sw} off.

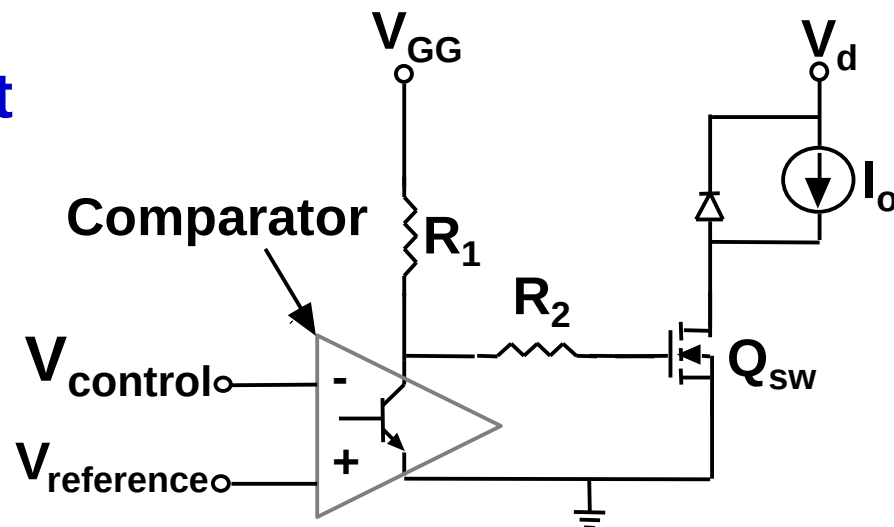


Fig. 28-2 Simple MOSFET gate drive circuit for low-speed and low-switching-frequency application

➤ Large $R_1 \rightarrow$ low losses, but large $R_1 \rightarrow R_2 \uparrow \rightarrow$ slow-down the MOSFET turn-on speed.



(b) Totem-pole gate drive circuit

Circuit operation

$V_{\text{control}} > V_{\text{reference}}$, comparator outputs high putting Q_{npn} on and thus Q_{sw} on.

$V_{\text{control}} < V_{\text{reference}}$, comparator outputs low putting Q_{pnp} on and thus Q_{sw} off.

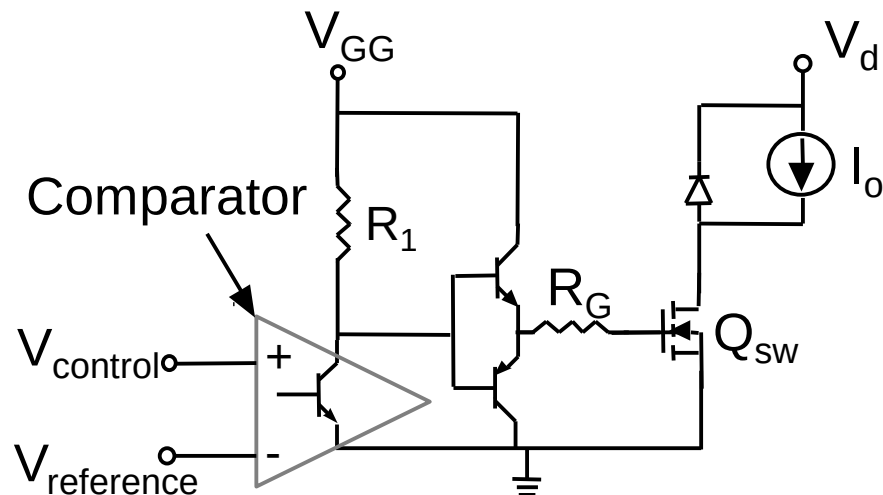
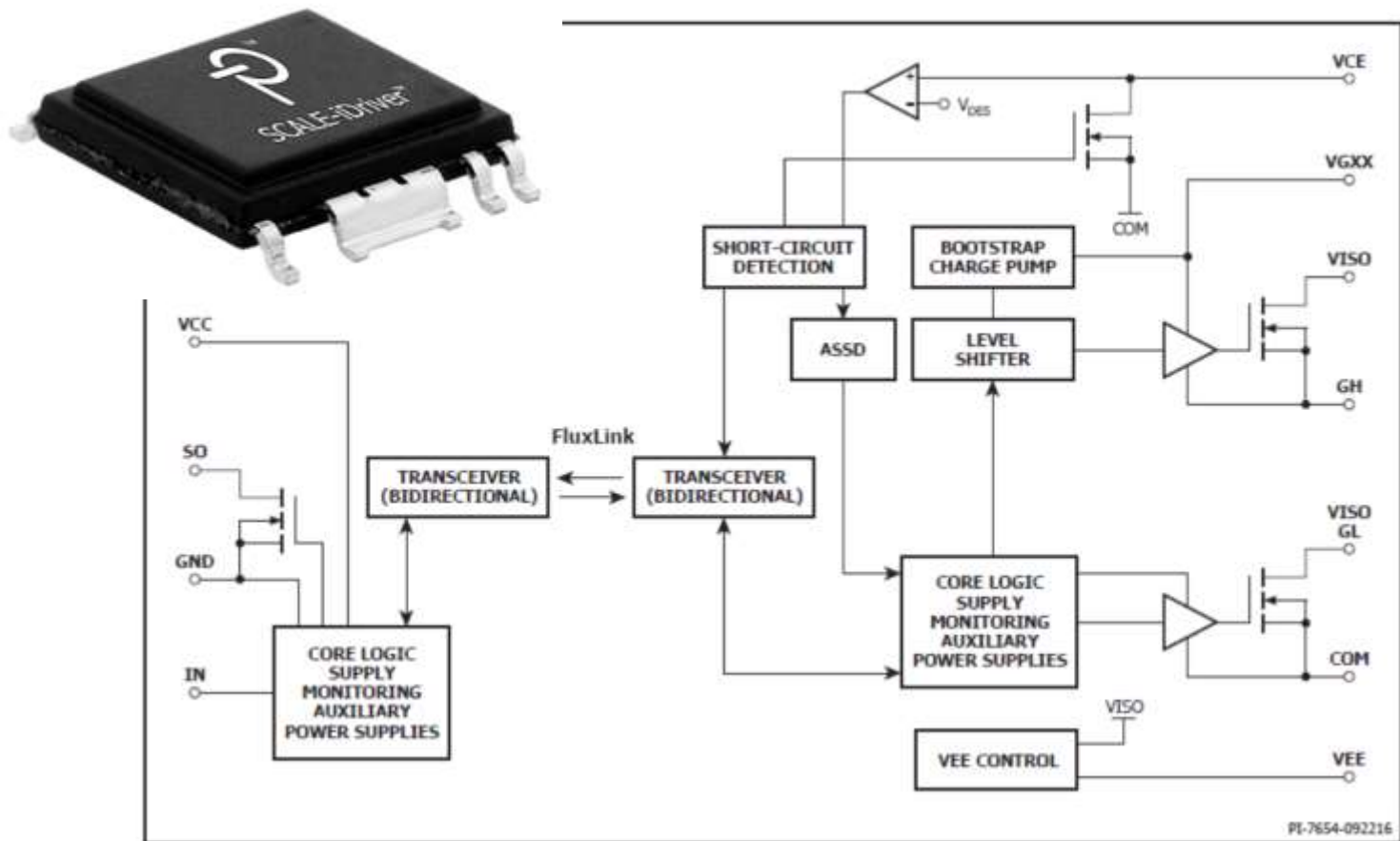


Fig. 28-3(a) Discrete totem-pole gate drive circuit

Since no steady-state current flows through R_G , R_G can be much smaller in value, which results in faster turn-on and turn-off times.



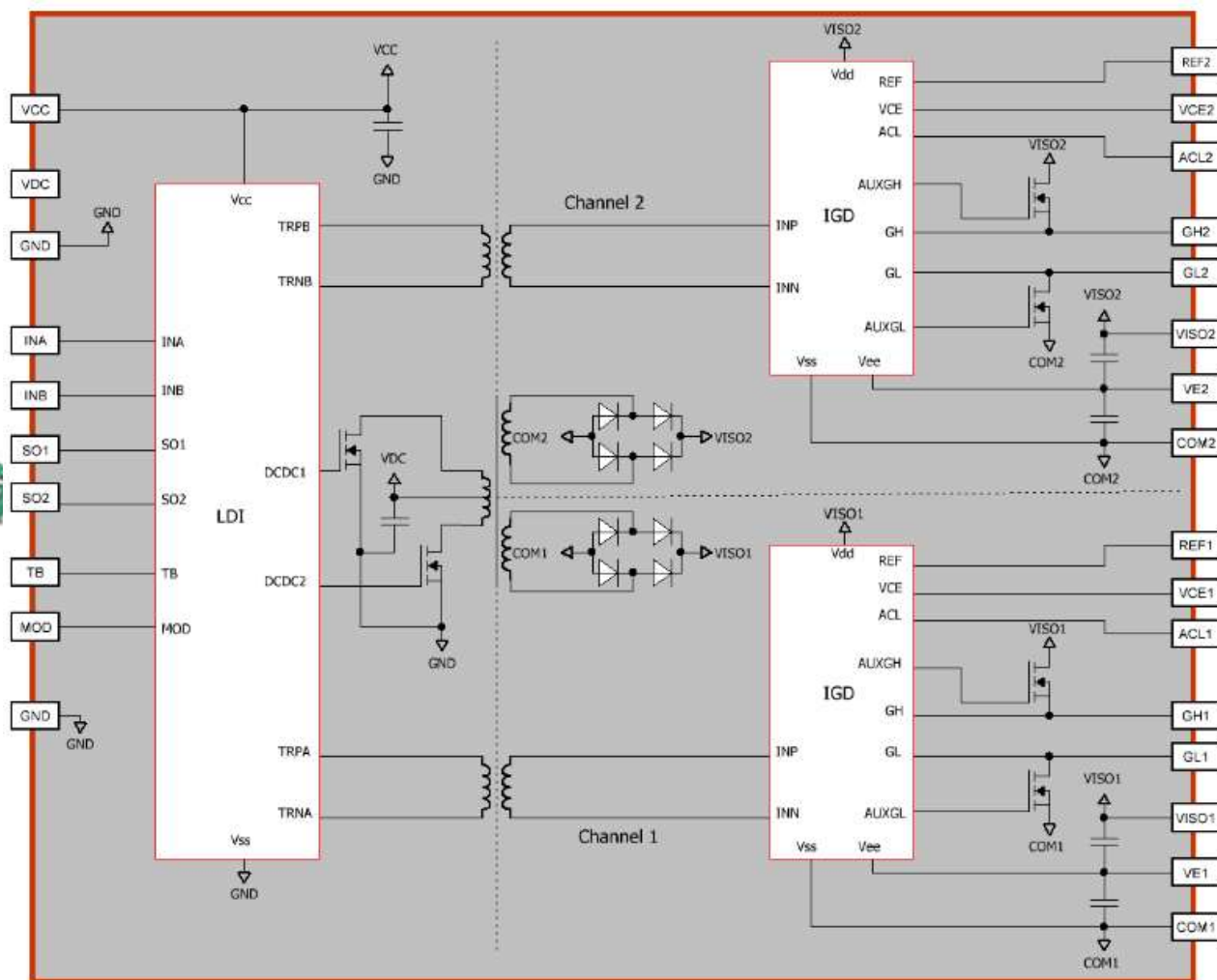
Example 5. Gate drive designed with integrated circuit or drive core



(a) Photo and block diagram of PI **SCALE-iDriver**

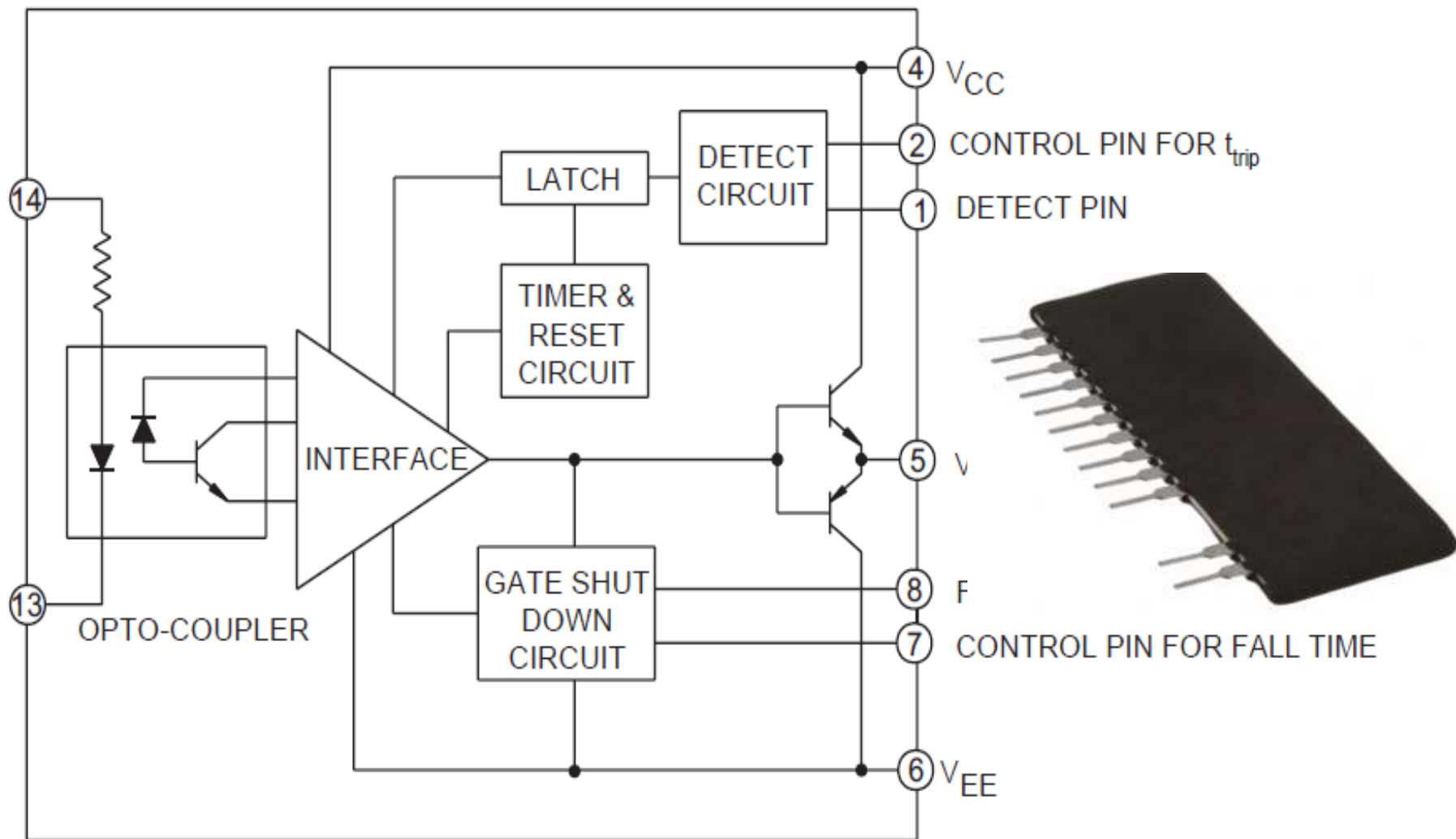


Photo of driver core



Block diagram of driver core

(b) Photo and block diagram of PI Driver core

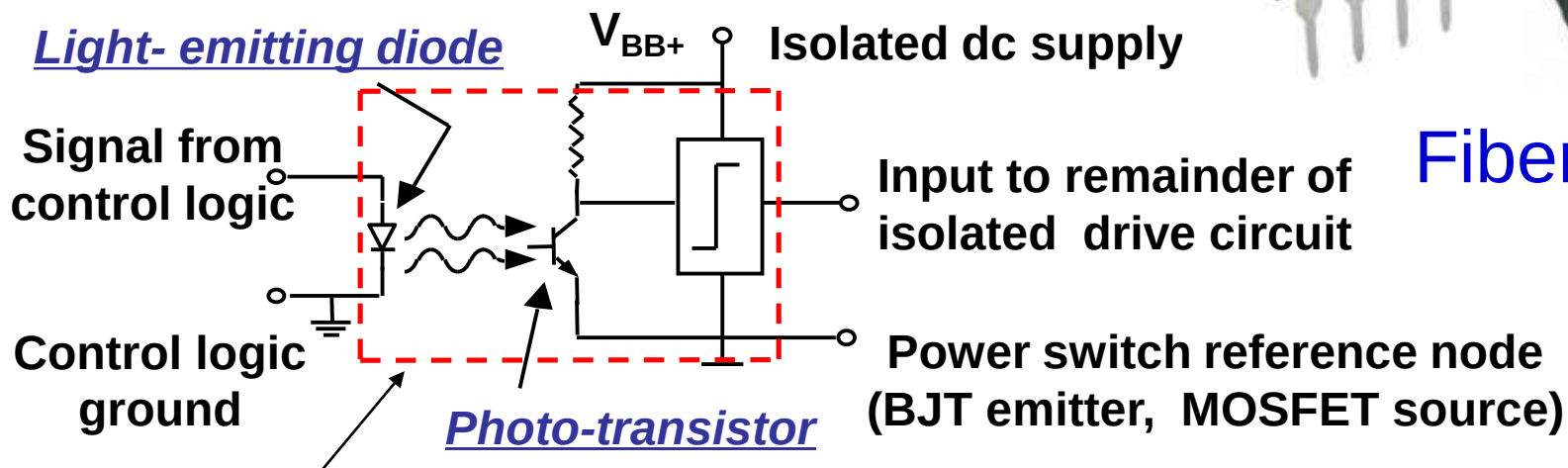


(b) Photo and block diagram of M57962 **Driver core**



(3) Electrical Isolation ways

(a) Opto-coupler (optical coupling) Fiber optics (optical coupling)

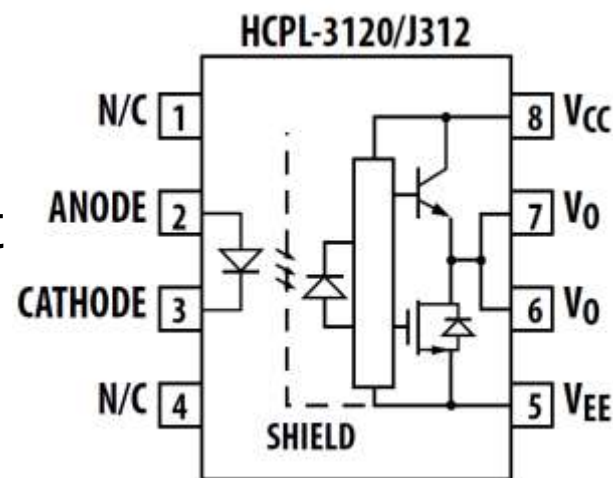


Opto-coupler Fig. 28-8

A positive signal from the control logic cause LED to emit light. The light falling on the base region of photo transistor and causes the photo transistor to turn on.

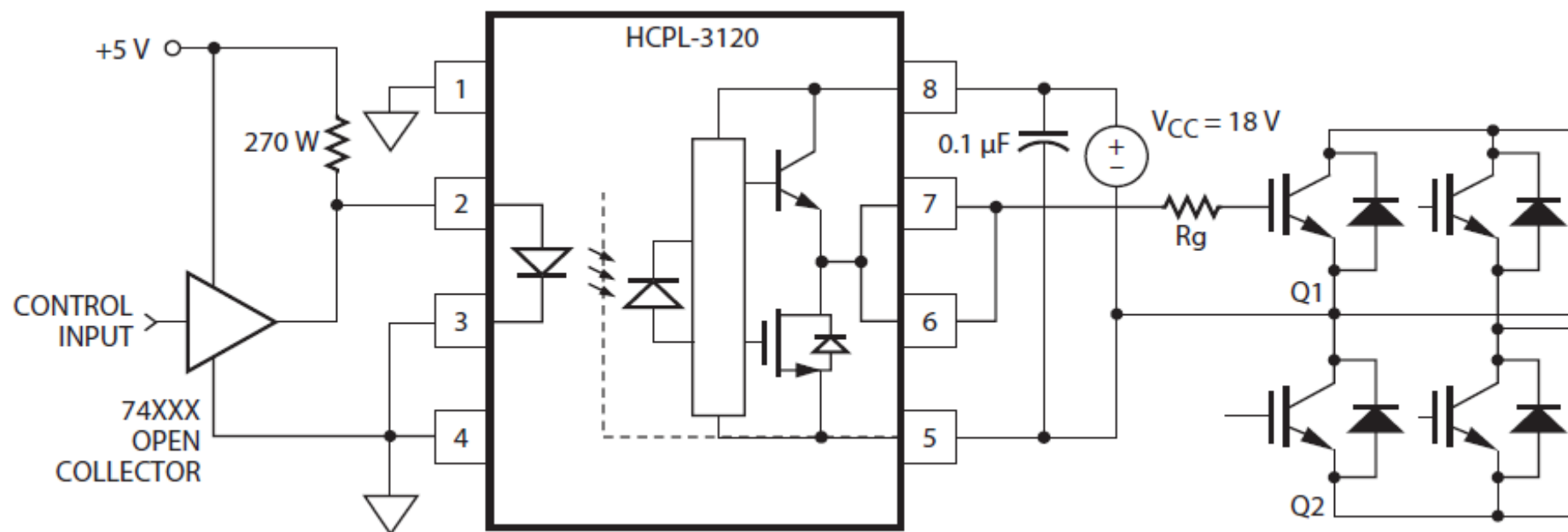


Fiber optics





Example 1. Gate drive with Opto-coupler isolation





(b) Inductive coupling

I. Pulse transformer coupling

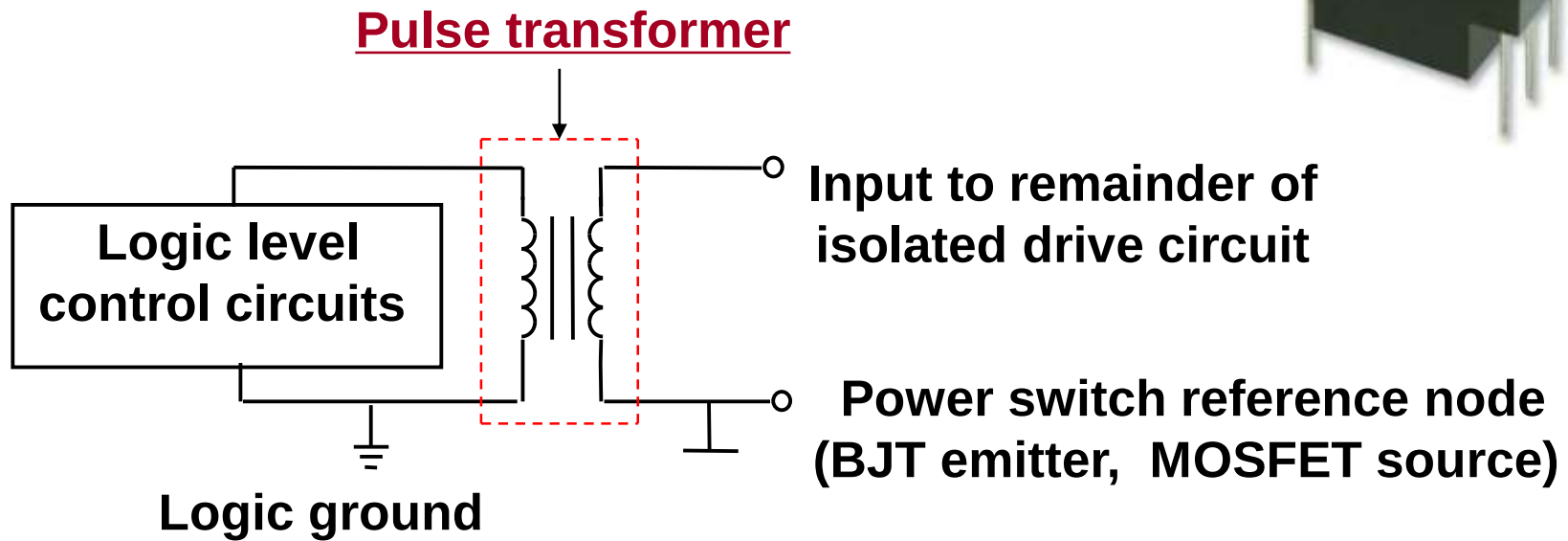


Fig. 28-9(a)

Applicable for high switching frequency (several tens of kHz or more) applications.



II . Coreless transformer

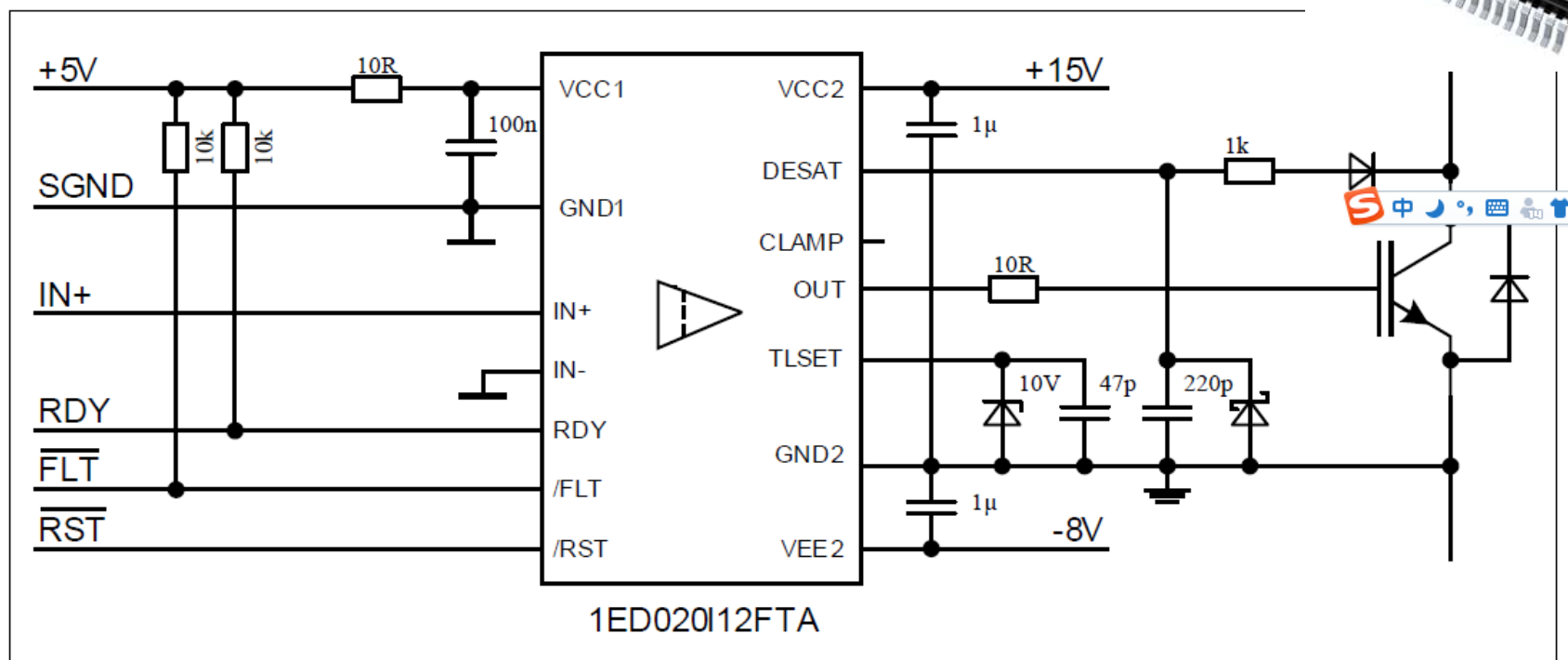
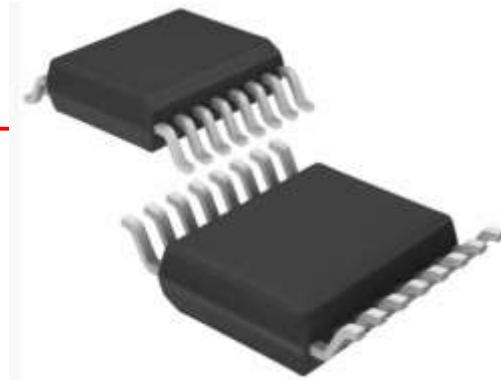
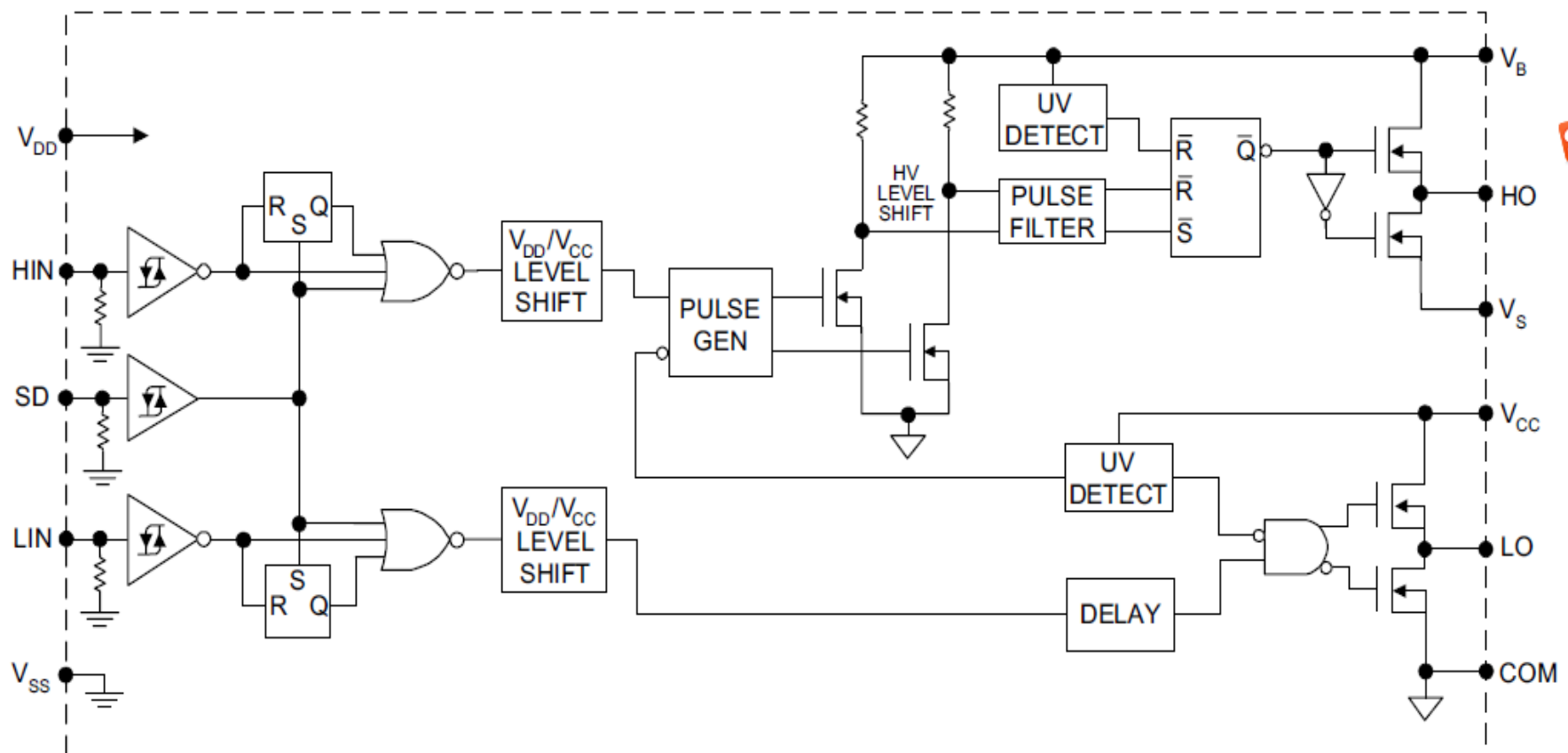


Photo and diagram of Infineon EiceDriver 1ED020I12FTA 29



(c) Level shifter (monolithic coupling) 电平移位器(单片耦合器)

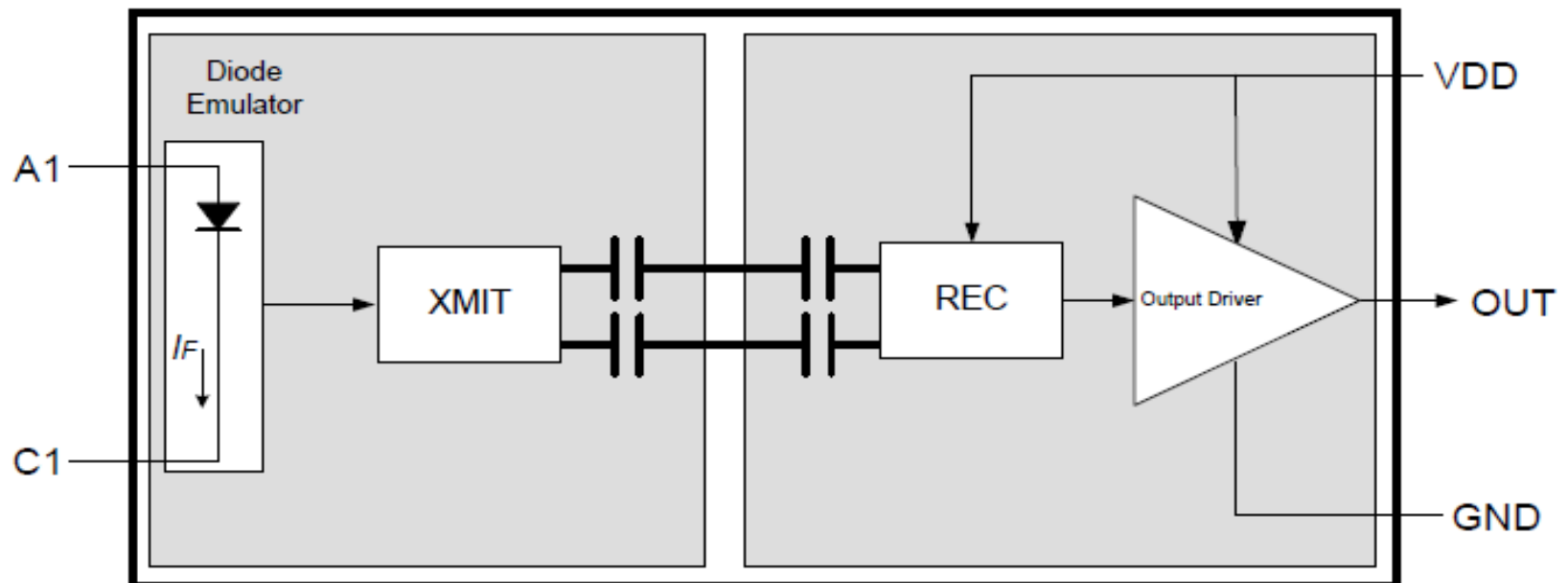
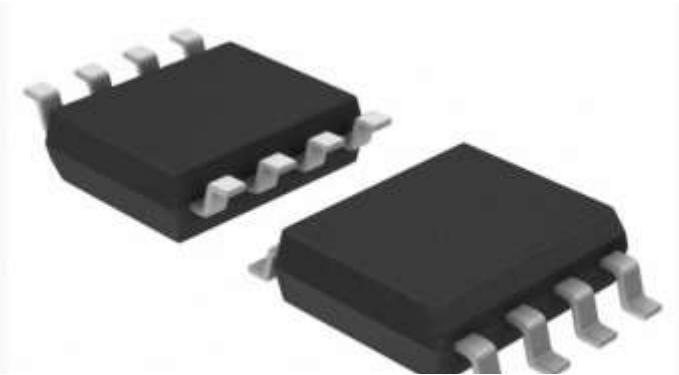
Example. IR2110 gate drive chip





(d) Capacitive coupler (capacitive coupling)

Example: Si8261 gate drive chip





Power devices **cannot** be protected against the **overcurrent by fuses** because they cannot act fast enough.

Normally way of providing protection is to **monitor collector-emitter saturation voltage V_{CE}** of a BJT , IGBT or V_{DS} of a MOSFET.

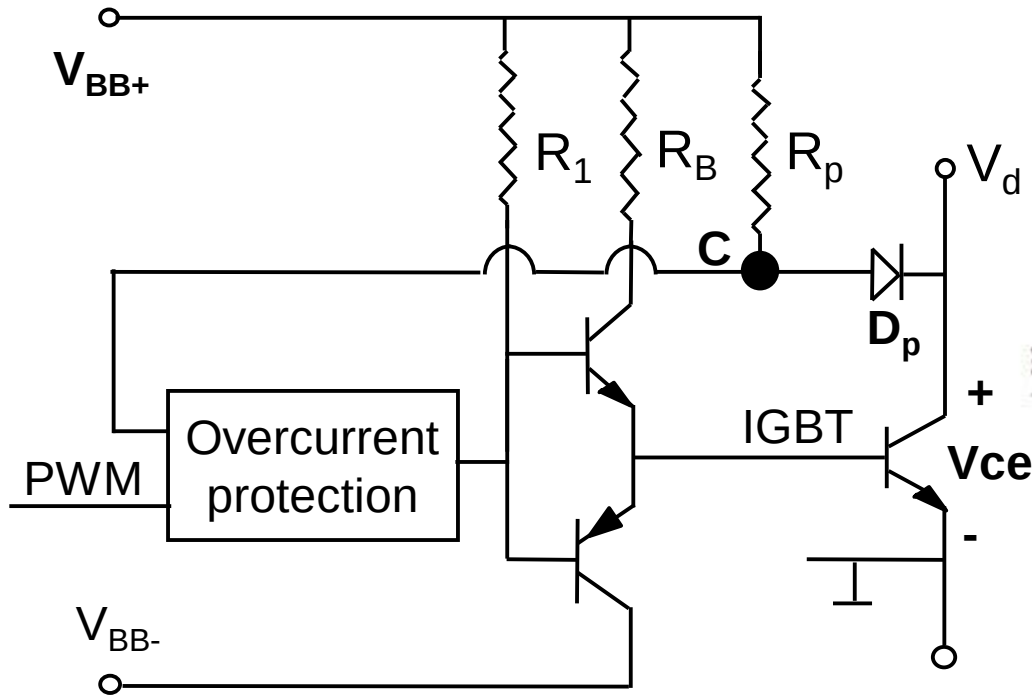
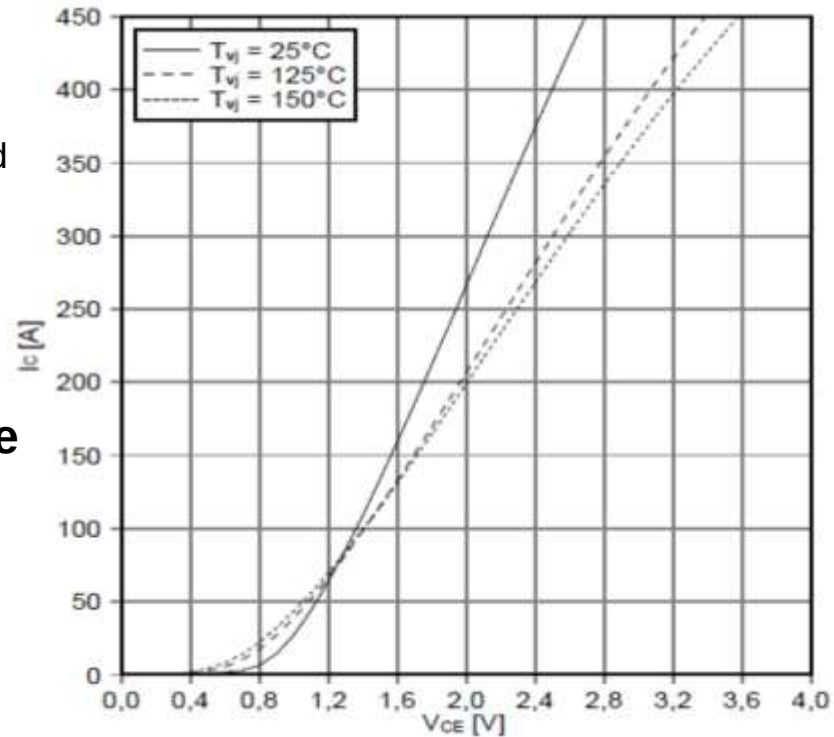


Fig. 28-23 (a)

(1) Analysis

Potential at node C equals diode drop $V_d + V_{CE(sat)}$ when IGBT is on. **Overcurrent will increase V_{CE}** and thus potential at node C.





(b) Overcurrent Protection ways

➤ Limiting Overcurrents by limiting gate drive current

Step-down converter with short circuit at $t = t_{sc}$

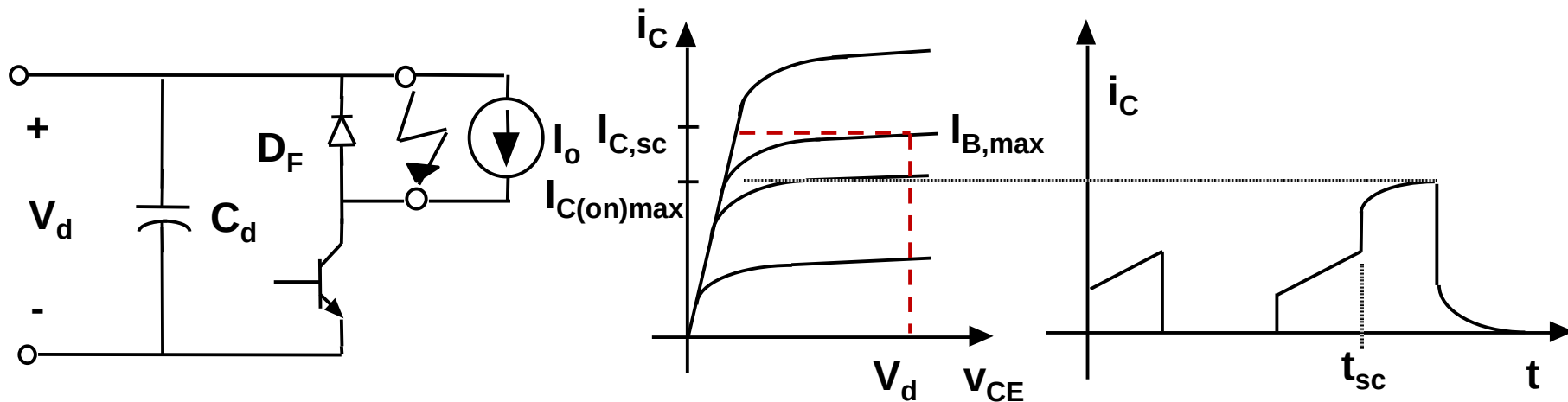


Fig 28-24 (a)

Fig. 28-24 (b) and (c)

Overcurrent limited to $I_{C(on)max} < I_{C,sc}$ by keeping $I_{B,max} < I_{C,sc}/\beta$

$I_{C,sc}$ = maximum allowable instantaneous collector current

Same approach can be used with **MOSFETs and IGBTs**.

➤ Switch should be **soft/slowly turned-off** when overcurrent occurs, especially short circuit.



(5) Design Considerations

(a) V_{GE} (V_{GS}) affects on-state losses and short circuit behavior:

$V_{GE+} \uparrow \rightarrow R_{dson}/V_{ce} \downarrow$, on-state loss $\downarrow$, short circuit $\uparrow$

Generally: $V_{GE} = 15V$

(b) Provide negative voltage V_{GE-} ($-V_{GS}$) reverse bias to turn off device

Generally: $V_{GE-} = -5V/-10V$

(c) Gate drive resistance R_G

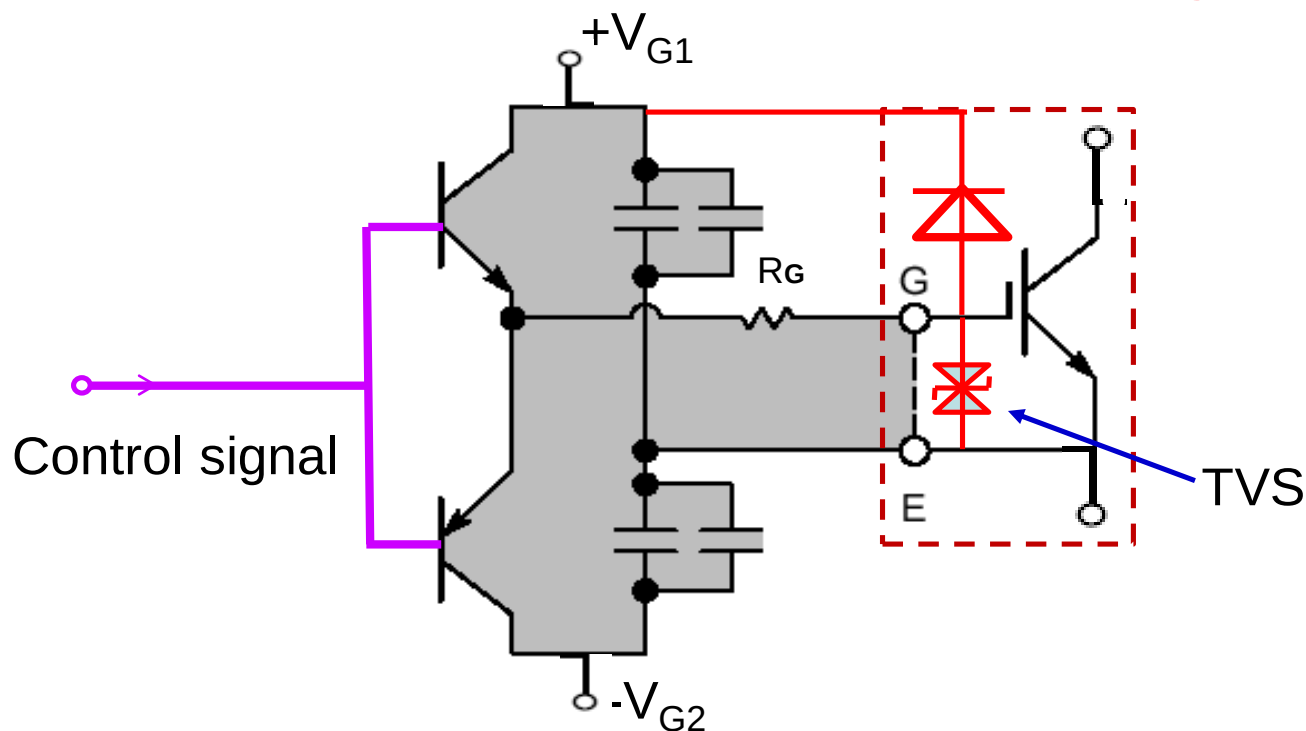
$R_G \downarrow \rightarrow$ transition speed of switches $\uparrow$, switching losses $\downarrow$, and vice versa. Too small R_G maybe cause unexpected resonant in gate.

Generally select: $R_G = 3 \sim 30(\Omega)$.



(d) Clamp of V_{GE} and protect V_{GE} avoid overvoltage by parallizingTVS diode between gate and source/collector

How to select the TVS break voltage?

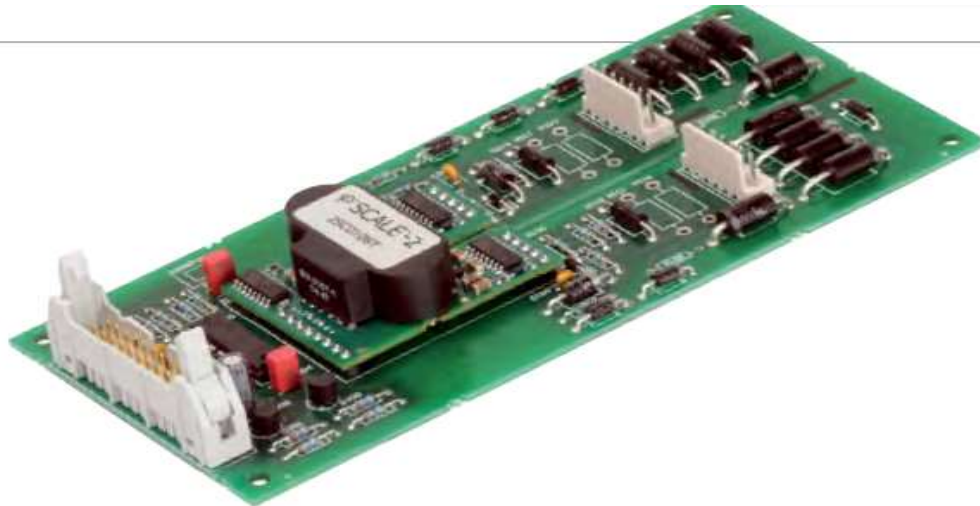


TVS should be placed closed to IGBT



(e) Circuit /Component Layout Consideration

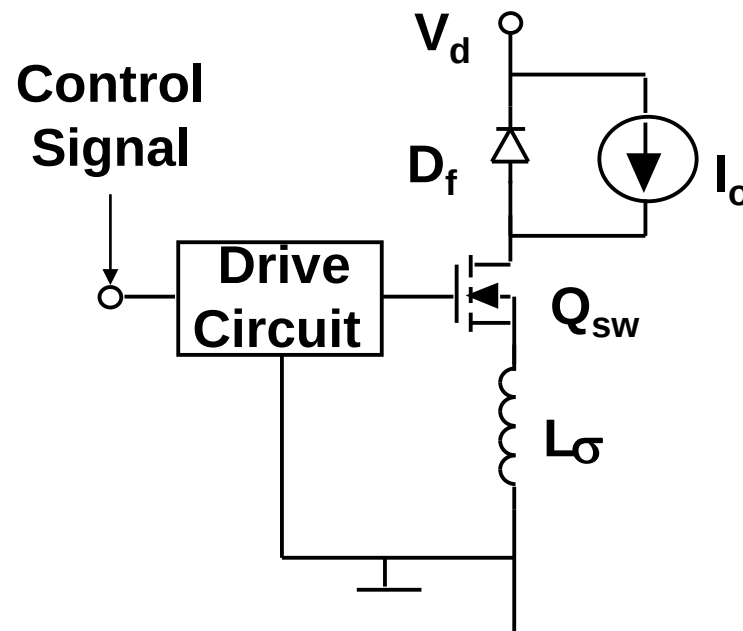
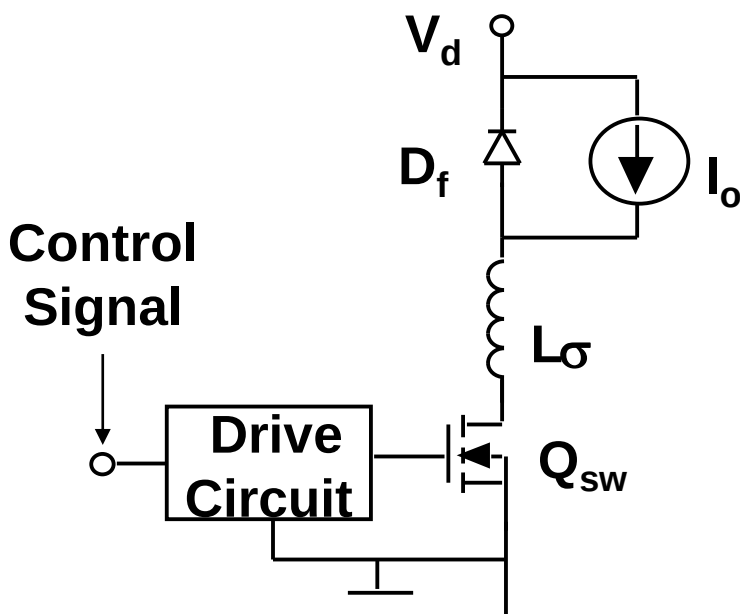
- ◆ **Minimizing stray inductance** in gate drive circuit loop.
Directly assemble driver board on IGBT module.
- ◆ Use **twist wire or shield conductor** to connect gate drive board to IGBT, MOSFET etc.
- ◆ Keep **enough distance** for electrical isolation.





1 Prime consideration is minimizing stray inductance

- Stray inductance in series with high-voltage side of power device Q_{sw} causes overvoltage at turn-off.
- Stray inductance in series with low-voltage side power device Q_{sw} can cause oscillations at turn-on and turn-off.



- One cm of unshielded conductor has about 5 nH of series inductance.
- Keep unshielded conductor lengths to an absolute minimum.



2 Use shielded conductors to connect drive circuit to power switch if there must be any appreciable separation (few cm or more) between them.

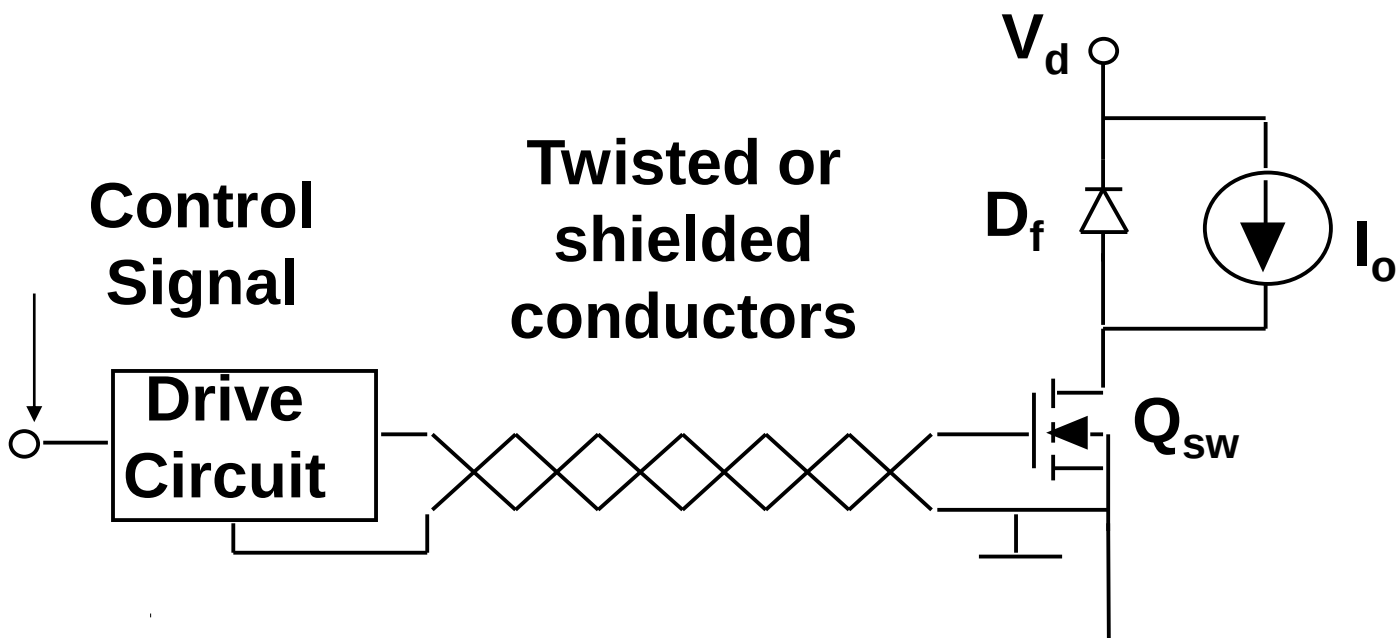


Fig. 28-27 (a)



Some power devices provided with four terminals, two input terminals and two power terminals, to short emitter connection to gate drive circuit, hence to minimize stray inductance in input circuit.

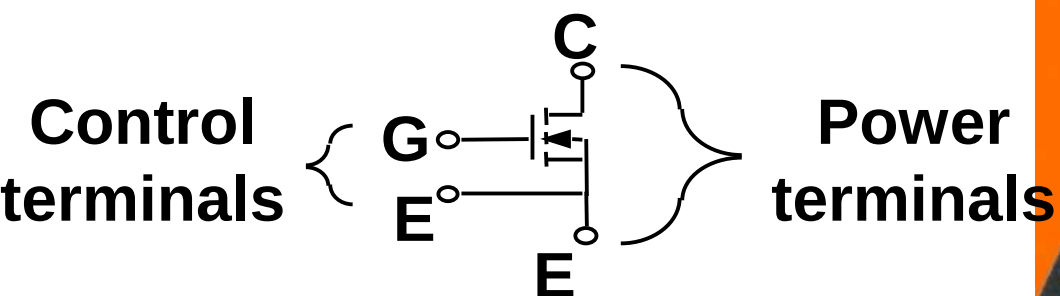


Fig. 28-27 (c)





Questions

- 1. What are the functions of Gate/Base drive circuits?**
- 2. What are the considerations in designing drive circuit?**
- 3. What are the advantages and disadvantages of transformer gate isolations?**
- 4. What are the advantages and disadvantages of opto-coupled gate isolations?**